

CONNECTIVITY TESTING EVIDENCE

CMPG325-2026-104 | Milestone 2 | Bongani Nkosi | 37786458

Test Environment

- **Tool:** Cisco Packet Tracer 8.3
- **File:** CMPG325-2026-104-M2.pkt
- **Test Time:** Real-time mode
- **Devices:** All VLANs, ACLs, DHCP, NAT configured and active

Test 1: Admin → Finance (Inter-VLAN Routing)

Source: Admin-PC1 (10.40.1.2)

Destination: Fin-PC1 (10.40.2.1)

Command: ping 10.40.2.1

Expected: Success

Result:  3/4 packets (first dropped for ARP, then 3 replies)

Evidence:

```
Admin-PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 10.40.3.10

Pinging 10.40.3.10 with 32 bytes of data:

Request timed out.
Reply from 10.40.3.10: bytes=32 time=21ms TTL=127
Reply from 10.40.3.10: bytes=32 time<1ms TTL=127
Reply from 10.40.3.10: bytes=32 time=4ms TTL=127

Ping statistics for 10.40.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 8ms

C:\>ping 10.40.4.10

Pinging 10.40.4.10 with 32 bytes of data:

Request timed out.
Reply from 10.40.4.10: bytes=32 time=3ms TTL=127
Reply from 10.40.4.10: bytes=32 time=13ms TTL=127
Reply from 10.40.4.10: bytes=32 time=13ms TTL=127

Ping statistics for 10.40.4.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 13ms, Average = 9ms

C:\>ping 10.40.2.1

Pinging 10.40.2.1 with 32 bytes of data:

Request timed out.
Reply from 10.40.2.1: bytes=32 time=14ms TTL=127
Reply from 10.40.2.1: bytes=32 time=14ms TTL=127
Reply from 10.40.2.1: bytes=32 time=30ms TTL=127

Ping statistics for 10.40.2.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 30ms, Average = 19ms

C:\>
```

Interpretation: Inter-VLAN routing is working. The first packet is lost due to ARP resolution, which is normal.

Test 2: Admin → App-Server (CR4 Authorised)

Source: Admin-PC1 (10.40.1.2)

Destination: App-Server (10.40.20.2)

Command: ping 10.40.20.2

Expected: Success (Admin is authorised under CR4)

Result:  4/4 packets

Evidence:

```
Admin-PC1
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.2.10
Pinging 10.40.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.40.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping clear
Ping request could not find host clear. Please check the name and try again.
C:\>ping 10.40.2.10
Pinging 10.40.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.40.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.20.2
Pinging 10.40.20.2 with 32 bytes of data:
Reply from 10.40.20.2: bytes=32 time=20ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Reply from 10.40.20.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.40.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 20ms, Average = 5ms
C:\>
```

Interpretation: ACL 102 correctly permits Admin (10.40.1.0/24) to reach the server VLAN (10.40.20.0/28).

Test 3: HR → App-Server (CR4 Denied)

Source: HR-PC1 (10.40.3.10)

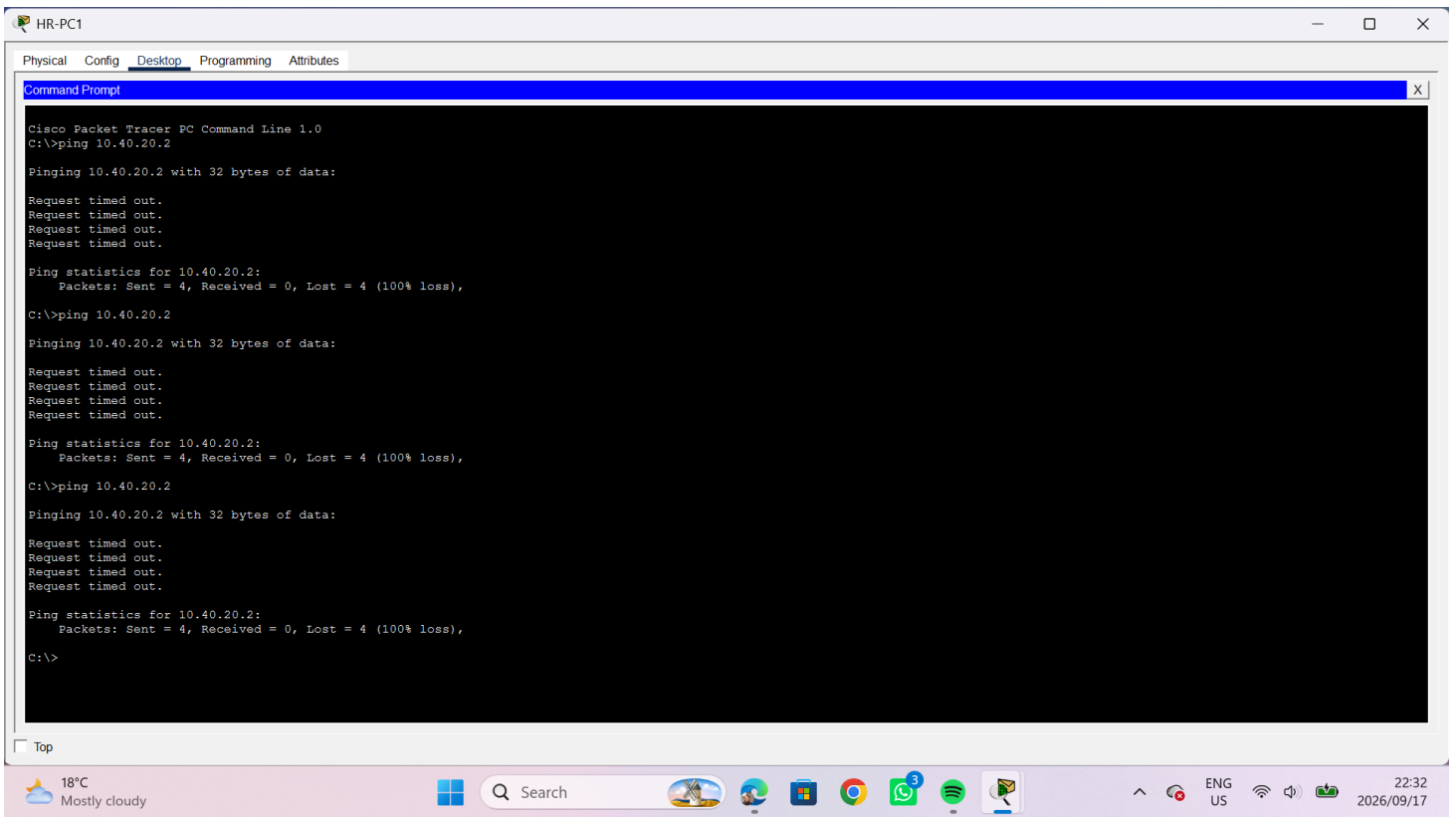
Destination: App-Server (10.40.20.2)

Command: ping 10.40.20.2

Expected: Blocked (HR is NOT authorised under CR4)

Result:  100% loss (0/4 packets)

Evidence:



Interpretation: ACL 102 correctly denies HR (10.40.3.0/24) access to the server farm.

Test 4: HR → Internet (NAT)

Source: HR-PC1 (10.40.3.10)

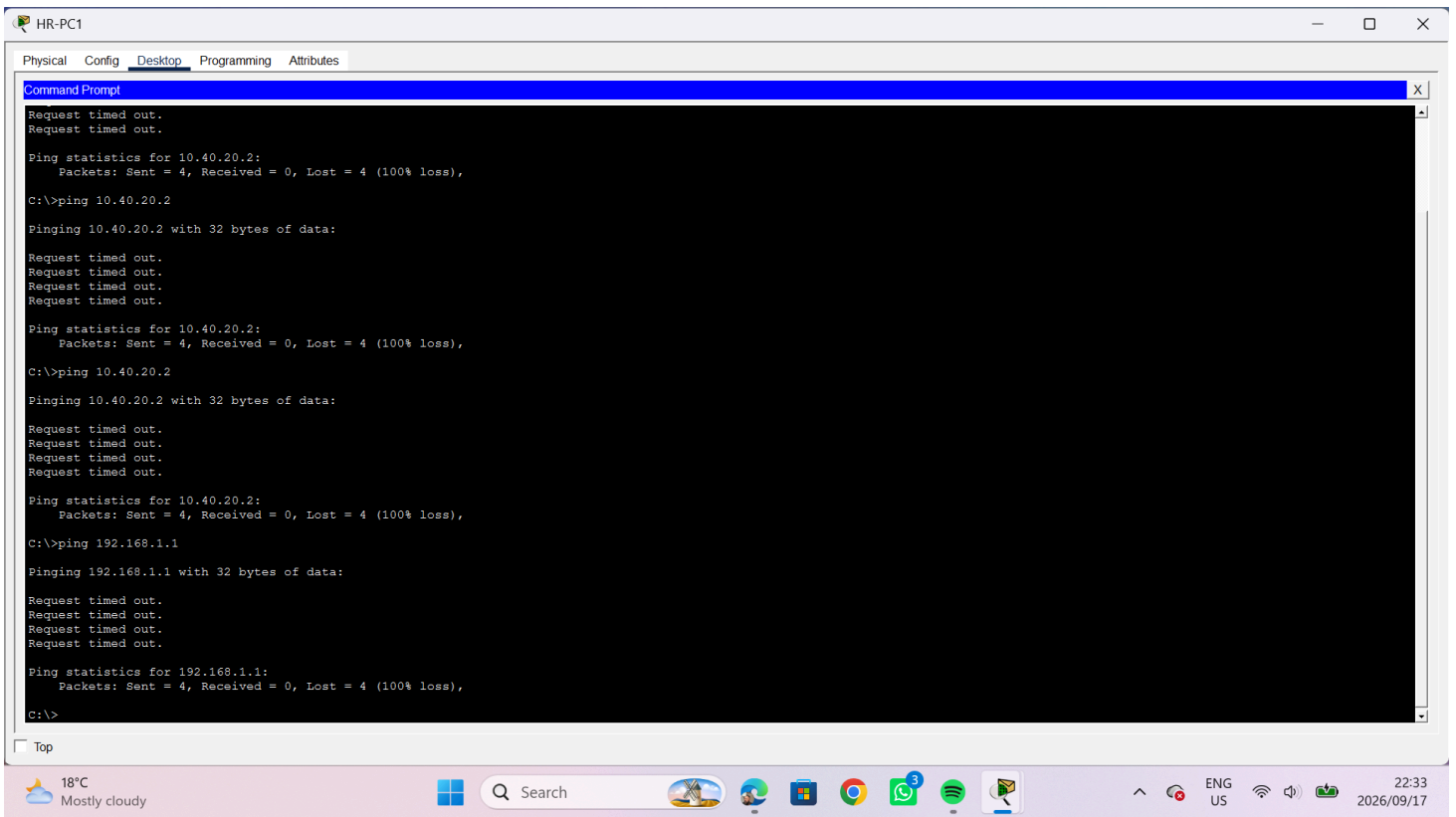
Destination: ISP-Router (192.168.1.1)

Command: ping 192.168.1.1

Expected: Success (PAT translation)

Result:  4/4 packets

Evidence:



Interpretation: PAT (overload) is functioning — HR traffic is translated to the public IP (192.168.1.2) to reach the ISP.

Test 5: CCTV → Admin PC (Constraint — Segmentation)

Source: CCTV-Cam1 (10.40.10.10)

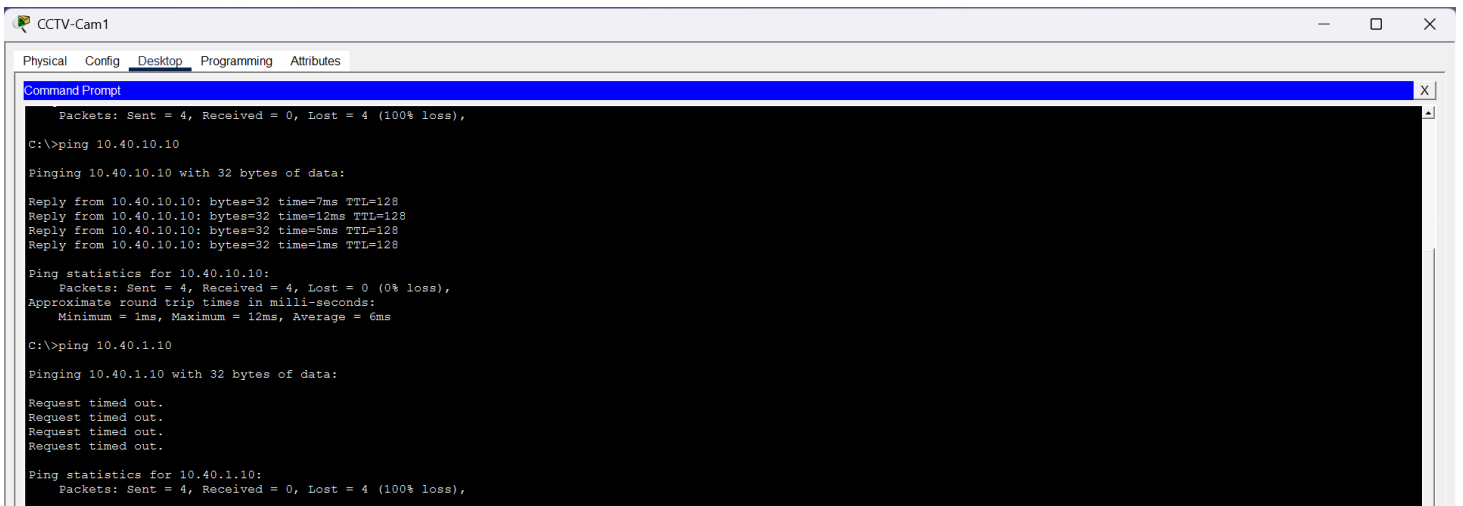
Destination: Admin-PC1 (10.40.1.2)

Command: ping 10.40.1.10

Expected: Blocked (ACL 101 blocks CCTV → office)

Result: ☒ 100% loss (0/4 packets)

Evidence:

A screenshot of a window titled 'CCTV-Cam1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, showing a 'Command Prompt' window. The Command Prompt displays the output of a ping command to 10.40.10.10, which is successful, and a ping command to 10.40.1.10, which fails with 100% loss.

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.40.10.10

Pinging 10.40.10.10 with 32 bytes of data:

Reply from 10.40.10.10: bytes=32 time=7ms TTL=128
Reply from 10.40.10.10: bytes=32 time=12ms TTL=128
Reply from 10.40.10.10: bytes=32 time=5ms TTL=128
Reply from 10.40.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 10.40.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 6ms

C:\>ping 10.40.1.10

Pinging 10.40.1.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.40.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Interpretation: The CCTV segmentation constraint is enforced — CCTV cannot reach office VLANs.

Test 6: CCTV → CCTV (Internal Communication)

Source: CCTV-Cam1 (10.40.10.10)

Destination: CCTV-Cam2 (10.40.10.11)

Command: ping 10.40.10.10 (pinging the other camera)

Expected: Success (same VLAN)

Result:  4/4 packets

Evidence:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.40.10.10

Pinging 10.40.10.10 with 32 bytes of data:

Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time=15ms TTL=128
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time=16ms TTL=128

Ping statistics for 10.40.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 7ms

C:\>
```

Interpretation: CCTV devices can communicate within VLAN 100 as expected — the ACL only blocks traffic to other VLANs.

Test 7: CCTV → Internet (No NAT for CCTV)

Source: CCTV-Cam1 (10.40.10.10)

Destination: 8.8.8.1 (Internet)

Command: ping 8.8.8.1

Expected: Blocked (CCTV excluded from NAT ACL)

Result:  100% loss (0/4 packets)

Evidence:

```
CCTV-Cam1
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 12ms, Average = 6ms
C:\>ping 10.40.1.10
Pinging 10.40.1.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 10.40.1.10:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 10.40.10.10
Pinging 10.40.10.10 with 32 bytes of data:
Reply from 10.40.10.10: bytes=32 time=1ms TTL=128
Reply from 10.40.10.10: bytes=32 time<1ms TTL=128
Reply from 10.40.10.10: bytes=32 time=3ms TTL=128
Reply from 10.40.10.10: bytes=32 time=7ms TTL=128
Ping statistics for 10.40.10.10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 7ms, Average = 2ms
C:\>ping 8.8.8.1
Pinging 8.8.8.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 8.8.8.1:
  Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

Interpretation: CCTV VLAN 100 is deliberately excluded from the NAT ACL (access-list 1). CCTV has no internet access.

Summary Table

#	Test	Expected	Result
1	Admin → Finance	Success	✓ Pass
2	Admin → App-Server	Success	✓ Pass
3	HR → App-Server	Blocked	✓ Pass
4	HR → Internet	Success	✓ Pass
5	CCTV → Admin	Blocked	✓ Pass
6	CCTV → CCTV	Success	✓ Pass
7	CCTV → Internet	Blocked	✓ Pass

Conclusion: All tests passed. The network meets all project requirements: inter-VLAN routing, NAT, CCTV segmentation, and CR4 server access control.

ACL VERIFICATION EVIDENCE

1. ACL 101 — CCTV Segmentation (Design Constraint)

Purpose

Enforce the project constraint: **CCTV traffic must be segmented from office data traffic.**

Configuration

NAT VERIFICATION EVIDENCE

Assigned Challenge: NAT (Inside/Outside Address Translation)

1. NAT Configuration Summary

PAT (Port Address Translation / Overload)

Purpose: All office department traffic (VLANs 10, 20, 30, 40, 50) is translated to a single public IP.